

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	Total(10)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I B.Lakshmi Lohitha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.lakshmi Lohitha

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

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Activity - CO2 AT1

1. Suppose that the data for analysis includes the attributes age. The age values for the data tuples are (in increasing order)
13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35,
35, 35, 35, 36, 40, 45, 46, 52, 70.

- what is the mean of the data? what is the median?
- what is the mode of the data? comment on the data's modality
- what is the midrange of the data?
- can you find (roughly) the first quartile (Q_1) and the third quartile (Q_3) of the data?
- Give the five-number summary of the data
- show a boxplot of the data.

a) Mean :- $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

$$= \frac{13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 + 22 + 25 + 25 + 25 + 25 + 30 + 33 + 33 + 35 + 35 + 35 + 35 + 36 + 40 + 45 + 46 + 52 + 70}{27}$$

$$= \frac{809}{27} = 30 \quad \therefore \boxed{\text{Mean} = 30}$$

Median :- $\frac{n+1}{2}$

$$\Rightarrow \frac{27+1}{2} = 14$$

The 14th value = 25

$$\boxed{\text{Median} = 25}$$

b) mode:- 25 & 35 are coming continuously

- As two values are repeating, the data is Bimodal.

c) midrange:-

$$\frac{\text{max} + \text{min}}{2} = \frac{70 + 13}{2}$$
$$= 41.5$$

$$\therefore \boxed{\text{midrange} = 41.5}$$

d) First Quartile (Q_1) & Third Quartile (Q_3)

lower half (first 13 values):-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

$\therefore$ middle value = 20

$$\boxed{Q_1 = 20}$$

Q_3 (upper half)

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

$\therefore$ median = 35

$$\boxed{Q_3 = 35}$$

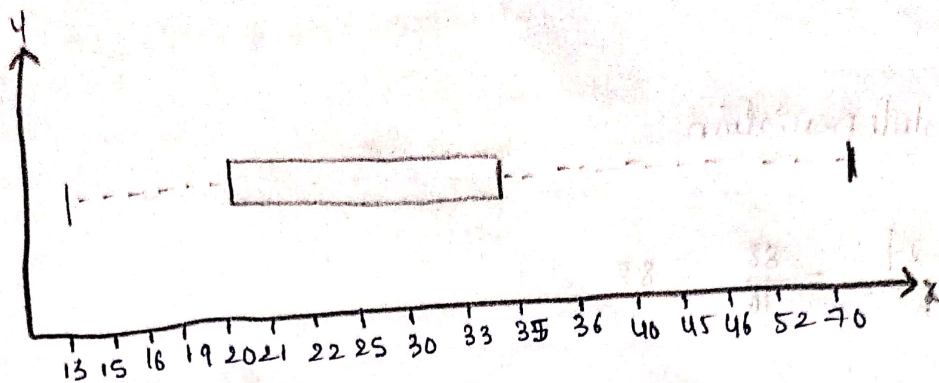
e) Five - number summary

• (min, Q_1 , median, Q_3 , max)

$\Rightarrow (13, 20, 25, 35, 70)$



f) box plot



2) Given the following measurements for the variable age:
18, 22, 25, 42, 28, 43, 33, 35, 56, 28, standardize the variable
by the following: compute the mean absolute deviation
of age.

Sol: mean = $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

$$\frac{18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28}{10}$$

$$= \frac{330}{10}$$

$$= 33$$

Absolute Deviation

$x(\text{age})$ $|x - \bar{x}|$

18 $|18 - 33| = 15$

22 $|22 - 33| = 11$

25 $|25 - 33| = 8$

42 $|42 - 33| = 9$

28 $|28 - 33| = 5$

43 $|43 - 33| = 10$

33 $|33 - 33| = 0$

35 $|35 - 33| = 2$

56 $|56 - 33| = 23$

28 $|28 - 33| = 5$

$$\rightarrow 15 + 11 + 8 + 9 + 5 + 16 + 10 + 2 + 23 + 5$$

$$= 88$$

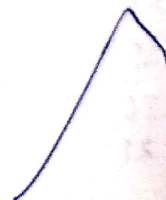
Mean Absolute Deviation

$$\frac{\sum |x - \bar{x}|}{n} = \frac{88}{10} = 8.8$$

3. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use something by bin means to smooth the above data. using a bin depth of 3. Illustrate your steps.

Sol: Bin depth = 3, No. of Bins = $\frac{\text{No. of values} - \text{range}}{\text{Bin depth}} = \frac{27}{3} = 9$

<u>Bin</u>	<u>values</u>	<u>Mean</u>	<u>smoothed values</u>
B ₁	13, 15, 16	$= (13 + 15 + 16) / 3 = 14.67$	14.67, 14.67, 14.67
B ₂	16, 19, 20	$= 18.33$	18.33, 18.33, 18.33
B ₃	20, 21, 22	$= 21$	21, 21, 21
B ₄	22, 25, 25	$= 24$	24, 24, 24
B ₅	25, 25, 30	$= 26.67$	26.67, 26.67, 26.67
B ₆	33, 33, 35	$= 33.67$	33.67, 33.67, 33.67
B ₇	35, 35, 35	$= 35$	35, 35, 35
B ₈	36, 40, 45	$= 40.33$	40.33, 40.33, 40.33
B ₉	46, 52, 70	$= 56$	56, 56, 56



4. Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000 (a) min-max normalization by setting min = 0 and max = 1 (b) z-score normalization.

Sol: a.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$\min = 200$$

$$\max = 1000$$

$$x' = \frac{x - 200}{800}$$

x	x'
200	$\frac{200 - 200}{800} = 0$
300	$\frac{300 - 200}{800} = 0.125$
400	$\frac{400 - 200}{800} = \frac{200}{800} = 0.250$
600	$\frac{600 - 200}{800} = \frac{400}{800} = 0.500$
1000	$\frac{1000 - 200}{800} = \frac{800}{800} = 1.000$

min-max normalization :- 0, 0.125, 0.250, 0.5, 1

b) $z = \frac{x - \mu}{\sigma}$

mean:

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$$

x	$x - \mu$	$(x - \mu)^2$
200	-300	90,000
300	-200	40,000
400	-100	10,000
600	100	10,000
1000	500	250,000

$$\sum (x - \mu)^2 = 4,00,000$$

$$\sigma = \sqrt{\frac{4,00,000}{5}} = \sqrt{80,000} = 282.84$$

$$\underline{\underline{Z\text{-score}}} = \frac{x - \mu}{\sigma}$$

<u>x</u>	<u>Z-score</u>
200	$-300/282.84 = -1.06$
300	$-200/282.84 = -0.71$
400	$-100/282.84 = -0.35$
600	$100/282.84 = 0.35$
1000	$500/282.84 = 1.77$

$\therefore (-1.06, -0.71, -0.35, 0.35, 1.77)$ - Z score

5. suppose a group of 12 sales price records has been sorted as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. partition them into three bins by each of the following methods.
- equal-frequency partitioning
 - equal-width partitioning
 - clustering.

Sol: Given,

Total records = 12

No. of bins = 3

a. equal frequency

Each bin contains $\frac{12}{3} = 4$ records



Bin	Values
B ₁	5, 10, 11, 13
B ₂	15, 35, 50, 55
B ₃	72, 92, 204, 215

b. equal width

1. calculate bin width $\frac{\text{max} - \text{min}}{\text{bin}} = \frac{215 - 5}{3} = \frac{210}{3} = 70$

create the bins:-

Bin 1: 5 - 74

Bin 2: 75 - 144

Bin 3: 145 - 215

Bin	Range	Values
B ₁	5 - 74	5, 10, 11, 13, 15, 35, 50, 55, 72
B ₂	75 - 144	92
B ₃	145 - 215	204, 215

c. clustering

cluster	value
C ₁	5, 10, 11, 13, 15, 35
C ₂	50, 55, 72, 92
C ₃	204, 215

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6. suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following results.

- calculate the mean, median and standard deviation of age and %fat
- draw the boxplots for age and %fat.
- draw a scatter plot and a q-q plot based on these two variables.
- normalize the two variables based on z-score normalization
- calculate the correlation co-efficient. Are these two variables positively (or negatively) correlated?

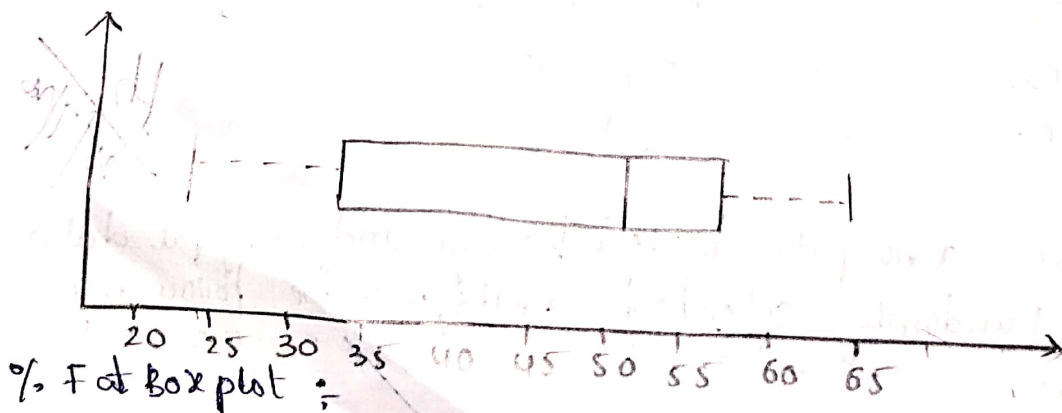
Sol:

	<u>age</u>	<u>%fat</u>
mean:	46.44	28.78
median:	51.00	30.70
standard deviation:	13.22	9.25

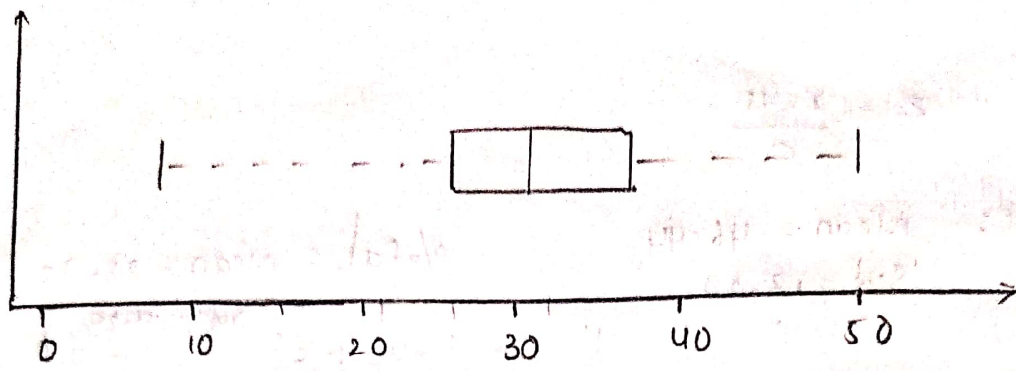
(b) Boxplot:

Age Boxplot (min, Q1, median, Q3, max)

(23, 33, 51, 57.5, 61)



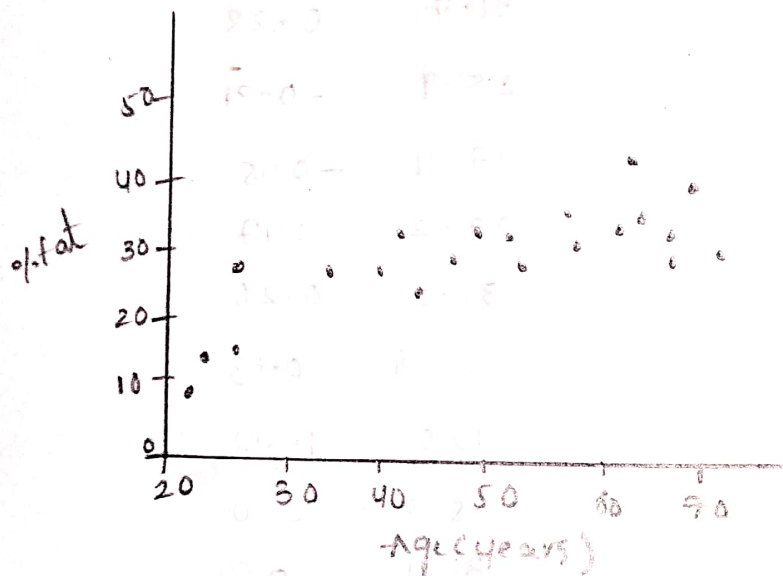
(7.8, 26.2, 30.7, 34.35, 42.5)



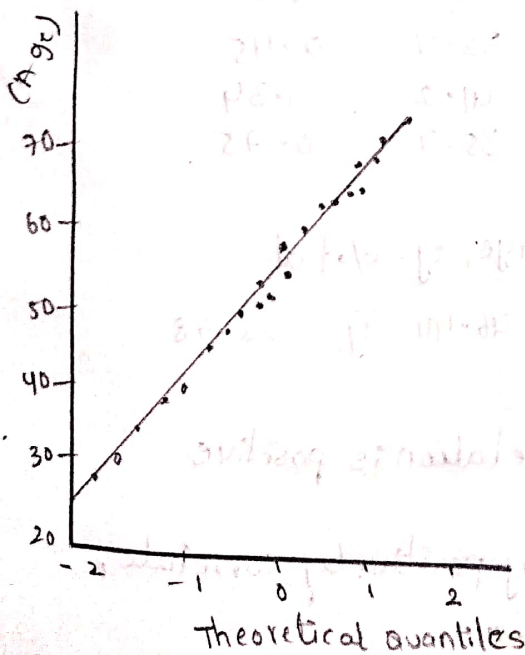
c. scatter plot

x-axis : Age

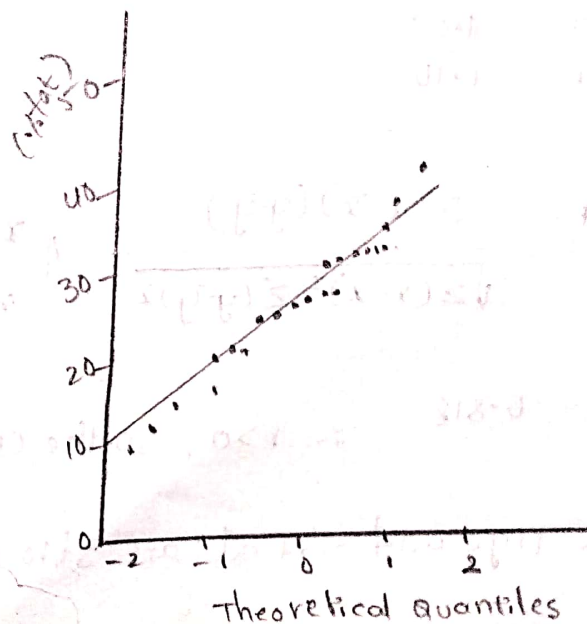
y-axis : %fat



Q-Q plot : Age



Q-Q plot : %fat



d. Z-score Normalization

$$Z = \frac{x - \mu}{\sigma}$$

Age:- mean = 46.44
S.d = 18.22

%fat:- mean = 28.78
S.d = 9.25

Age Z-score

23 -1.77

23 -1.77

27 -1.47

27 -1.47

39 -0.56

41 -0.41

47 0.04

49 0.19

50 0.27

52 0.42

54 0.57

54 0.57

56 0.72

57 0.80

58 0.87

58 0.87

60 1.03

61 1.10

%fat Z-score

9.5 -2.08

26.5 -0.25

27.8 -2.27

17.9 -1.19

31.4 0.28

25.9 -0.31

27.4 -0.15

27.2 -0.17

31.2 0.26

34.6 0.63

42.5 1.48

28.8 0.00

33.4 0.50

30.2 0.15

34.1 0.57

32.9 0.45

41.2 1.34

35.7 0.75

$$e. \quad r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

, $x = \text{Age}$, $y = \% \text{fat}$

$\bar{x} = 46.44$, $\bar{y} = 28.78$

$r = 0.818$

$\therefore r > 0$, so the correlation is positive

$\therefore$ Age and %Fat are strongly positively correlate.

